Guruprasad H D

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GitHub | LinkedIn

Professional Summary

Computer Science postgraduate with a strong foundation in **C/C++, Data Structures and Algorithms, Operating Systems, Computer Networks, Linux, and software development**. Hands-on experience developing software systems using **Java, Python, Spring Boot, Django, REST APIs, SQL, AWS, and MQTT**. Experienced in debugging, testing, API development, database-backed applications, and IoT data pipelines. Interested in **system-level software, embedded systems, networking protocols, wireless communication, and firmware development**.

Education

University Visvesvaraya College of Engineering (UVCE) *2025 – Present*
M.Tech in Computer Science and Engineering Bengaluru, Karnataka

Government Engineering College, Hassan *2021 – 2025*
B.E. in Computer Science and Engineering – CGPA: 8.80/10 Hassan, Karnataka

Technical Skills

- **Programming Languages:** C, C++, Java, Python, SQL, JavaScript
- **Core Programming:** Data Structures and Algorithms, Object-Oriented Programming, Pointers, Memory Management, Problem Solving
- **Systems & Networking:** Operating Systems, Computer Networks, TCP/IP, Wi-Fi Fundamentals, MQTT, Embedded Systems Fundamentals
- **Operating Systems & Tools:** Linux, Windows, Git, GitHub, Postman, VS Code
- **Frameworks:** Spring Boot, Django, Django REST Framework, React
- **Databases:** MySQL, PostgreSQL, SQLite, DynamoDB
- **Cloud & IoT:** AWS IoT Core, AWS Lambda, DynamoDB, MQTT

Experience

Parvam Software Solutions *Feb 2025 – Oct 2025*
Software Engineering Intern Onsite

– Developed and maintained software applications using **Java, Spring Boot, Python, SQL, and REST APIs**.
– Designed backend components and database operations using **Spring Boot, JPA/Hibernate, and MySQL**.
– Implemented RESTful APIs and integrated frontend components with backend services.
– Performed application **debugging, testing, troubleshooting, and performance optimization** to improve software reliability.
– Used **Git and GitHub** for source control and collaborated during development and testing.

Projects

AWS IoT Environmental Monitoring and Analytics *2026*
AWS IoT Core, MQTT, AWS Lambda, DynamoDB, Python

– Developed an IoT system for collecting and processing **real-time environmental sensor telemetry**.
– Used the lightweight **MQTT publish/subscribe protocol** to transmit temperature and humidity data from IoT devices.
– Configured **AWS IoT Core** for device communication and MQTT message handling.
– Implemented event-driven processing using **AWS Lambda** and stored device telemetry in DynamoDB.

AI-Powered Rail Complaint Management System *2025*

Python, Django, REST APIs, OpenCV, NLP, SQL

– Designed a modular web-based complaint management system using **Python, Django, REST APIs, and SQL**.

– Implemented backend logic for complaint registration, processing, classification, and database persistence.

– Integrated NLP and image-processing components for automated complaint analysis.

– Performed debugging and testing across backend, database, and application layers.

WealthManager – Portfolio Analytics Platform *2025*

React, Django REST Framework, PostgreSQL, Chart.js

– Developed a full-stack portfolio analytics application using a modular **frontend-backend architecture**.

– Designed REST APIs for portfolio holdings, allocation, performance, and financial summaries.

– Implemented database-backed services and integrated API responses with React-based visualization components.

– Debugged API integration, deployment, and runtime issues across the application stack.

TipEase – Digital Payment Platform *2024*

Django, Python, Razorpay API, Bootstrap, SQLite

– Developed a secure web application for digital tipping and payment processing.

– Integrated the **Razorpay payment API** with backend transaction-processing workflows.

– Implemented authentication, database operations, and responsive user dashboards.

Relevant Coursework

Data Structures and Algorithms **Operating Systems** **Advanced Operating Systems** **Computer Networks** Database Management Systems Object-Oriented Programming Internet of Things

Achievements

- Solved **250+ coding problems** covering data structures, algorithms, and problem-solving concepts.
- **Finalist – IISF Hackathon 2024**, developing a technology-based solution under competitive constraints.
- **Finalist – TechForSociety Hackathon 2025**, working collaboratively on a software solution for a real-world problem.
- Built multiple software, cloud, IoT, and full-stack projects available through GitHub.

Certifications

AWS Cloud Practitioner Essentials | Python Programming | SQL